

HIP DOCUMENTATION

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September 29

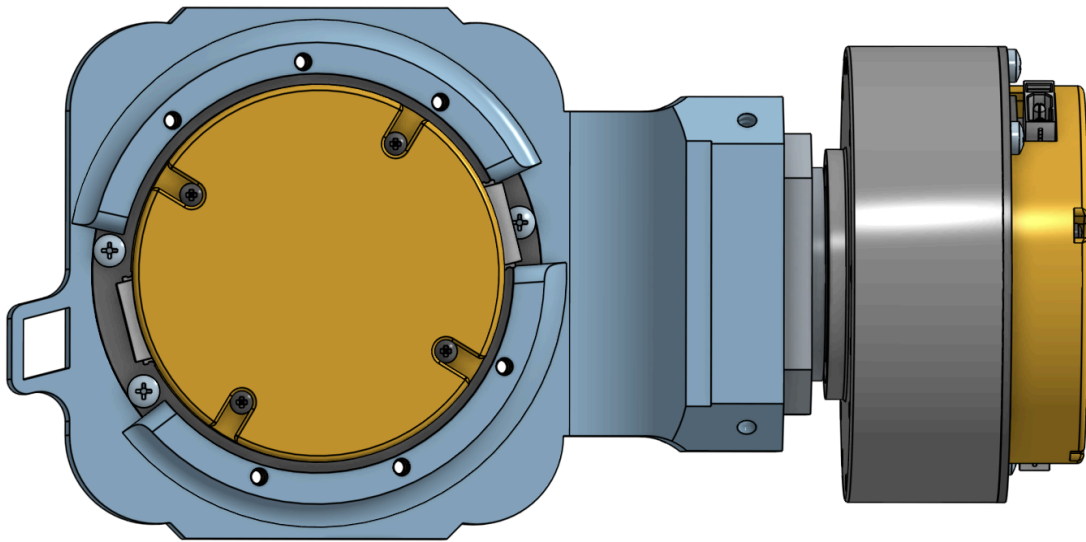
- Began brainstorming the redesign for the hip mount
- Requirements:
 - Secure connection between the body and the hip
 - The motor must be securely fastened to a mount
 - Extend outside the body to reduce the chances of contact between the hip and the body when turning

October 1

- Constructed a rough CAD prototype for the motor mount
 - Totally encapsulated the motor
 - The hip was extended out to avoid contact when rotating

October 6

- Created points of contact between the motor and motor mount
- Refined the shape of the mount and added a slot for wiring



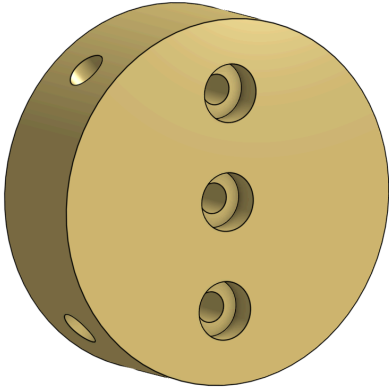
October 15

- Brainstormed ways to have a more secure connection within the hip and mount.

- Came up with a “puck” design that would go inside the hip
 - Screw running through the motor mount & hip, then the puck
 - Bolt the puck down to the motor mount or hip to make it more secure

October 22

- Continued work on the “puck” design
- Made sure the joints had sufficient tolerance levels to reduce the possibility of parts colliding
- Began adding screws to fix the motor to the hip joint

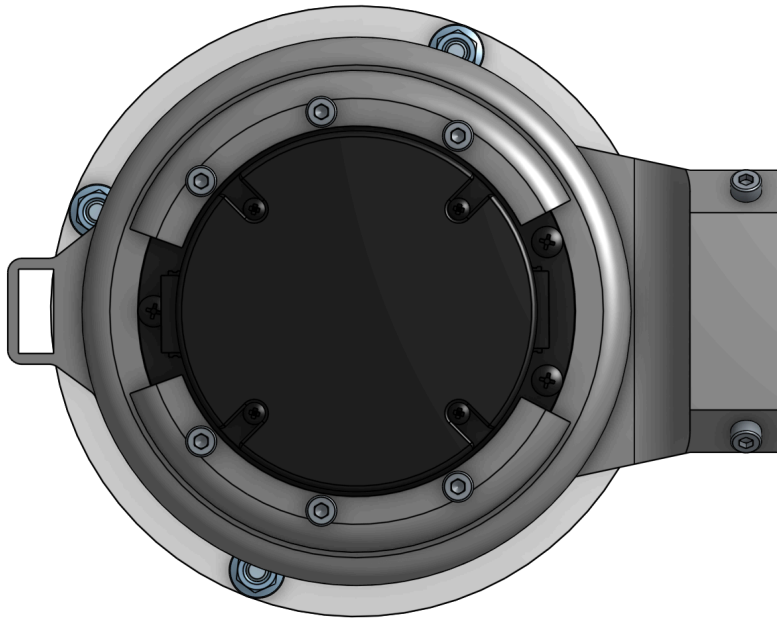


October 27

- Imported the new hip file into the main Quadrupe document
- Bolted puck to the motor mount

November 12

- Slimmed down motor mount
 - Cuts down on materials
 - Reduces weight
 - Sleeker design

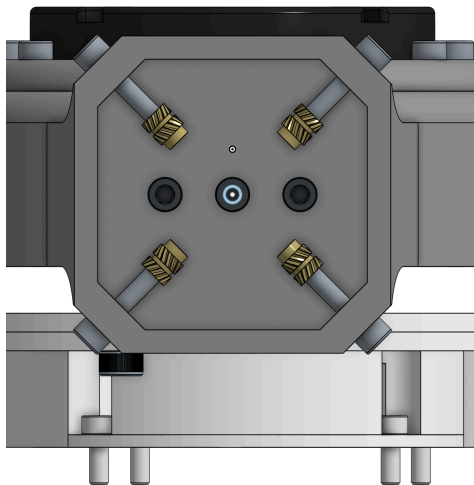


November 17

- Assembled a prototype of the printed hip and leg for test fitment
- Found some mistakes/areas to improve
- Reprinted the new parts and retested the fitment

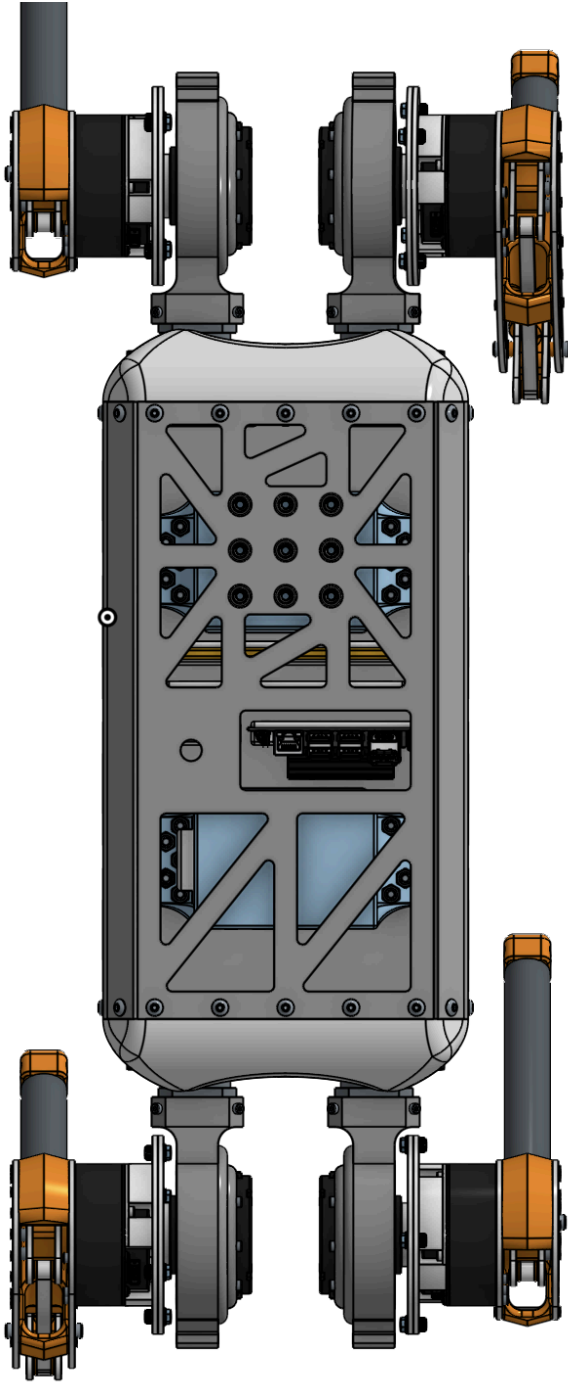
November 19

- Added threaded inserts into all holes in 3D-printed parts



December 1

- Full robot assembly completed



December 3

- 3D-printed the hip out of PETG, tuned the printer settings, such as temperature, to optimize the print quality
 - Prints still came out flawed, need to further optimize settings to increase print quality

December

January